

E-commerce sales

1. Dataset Overview

Start by understanding the dataset. Typically, e-commerce sales data will have several columns, such as:

- **Order ID:** Unique identifier for each order.
- **Customer ID:** Unique identifier for each customer.
- **Product Category:** Categories of products sold (e.g., electronics, clothing, etc.).
- **Price:** The cost of a single product.
- **Quantity:** Number of items purchased in each order.
- **Total Sales:** Price * Quantity.
- **Order Date:** Date when the order was placed.
- **Shipping Info:** Where the item was shipped, shipping duration, etc.
- **Discounts/Offers:** Any discounts applied to the order.

Steps to follow:

1. **Data Cleaning:** Ensure no missing values, handle duplicates, and convert date fields properly.
 - Identify missing values and decide whether to drop or impute them.
 - Correct any discrepancies in formats.
2. **Data Exploration:** Understand the relationships between variables.
 - Plot sales over time to see trends.
 - Calculate the average order value, total sales per customer, and products sold per category.

2. Exploratory Data Analysis (EDA)

Perform EDA to uncover insights from the dataset. Here's what you should focus on:

- **Sales Trends:** Plot sales over time to observe patterns, seasonality, or outliers.
- **Customer Insights:** Identify high-value customers based on frequency and monetary value (RFM analysis).
- **Product Insights:** Analyze the best-selling and least-selling products.
- **Geographical Analysis:** If you have shipping info, analyze which regions generate the most revenue.
- **Discounts and Offers:** Evaluate how discounts affect sales.

Methods for EDA:

- Use libraries like **pandas** for data manipulation.
- Use **matplotlib** or **seaborn** for visualizations (e.g., bar charts, histograms, and time-series graphs).
- Calculate **summary statistics** (mean, median, mode, etc.) for numerical columns.
- Create pivot tables to explore sales by category, region, and customer.

◆ Visualize the Data

Make charts! People understand graphs better than numbers.

Use:

- **Bar charts** to show top products
- **Line charts** to show sales over time
- **Pie charts** to show market share
- **Heatmaps** to show correlations between variables

3. Summary of Findings

Based on your analysis, summarize the main insights:

- **Sales Trends:** Highlight any increase or decrease in sales over time (e.g., spikes during holidays).
- **Top Products:** List the most sold products by quantity and revenue.
- **Customer Segments:** Identify if there are specific customer types that generate more revenue.
- **Geographic Insights:** Mention the regions that perform the best.
- **Discount Impact:** Determine if discounts lead to increased sales or not.

4. Recommendations

Based on the data insights, you can suggest actions:

- **Promotions:** Suggest targeted promotions for the best-selling products.
- **Customer Retention:** Propose a loyalty program for high-value customers.
- **Stock Management:** Recommend improving stock for high-demand products.
- **Price Adjustments:** If discounts are affecting the average sale price, advise on pricing strategy adjustments.

◆ Libraries Used

- `pandas`: For data manipulation
- `seaborn` and `matplotlib.pyplot`: For visualization

◆ Current Operations Performed

Operation	Description
<code>files.upload()</code>	Uploading a CSV file into the Colab session
<code>pd.read_csv()</code>	Reading the uploaded file into a DataFrame
<code>dataset.head(5)</code>	Displaying the first 5 rows
<code>dataset.describe()</code>	Getting statistical summary of numerical columns
<code>dataset.info()</code>	Checking data types and non-null counts
<code>dataset.isnull().sum()</code>	Finding missing values in each column